UQFF

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$n = [int]# UQFF  "PAPER_{0:D3}" -f [int]# UQFF
$n = [int]# UQFF PAPER_276
```

Andromeda Friedmann-UQFF Gravity Coupling: H(z)×t Expansion Term and H_UQFF Near-Unity Resonance

Author: Daniel T. Murphy
Session: 76 (March 2026)
Module: ANDROMEDA_UQFF_MODULE.cpp (M31 Master, UQFF 2.0)
WOLFRAM_TERM: `AndromedaUQFF:g_exp=G*M/r^2*H(z)*t; H(z)=H0*Sqrt[Om_m*(1+z)^3+Om_L]/`
`H_UQFF=H(z)*t_H~0.987 [PAPER_276]`

Abstract

This paper introduces a new UQFF gravity term for the Andromeda Galaxy (M31): the **Friedmann-UQFF Expansion Coupling**, defined as g_expansion = $(G\cdot M/r^2) \times H(z) \times t$. For Andromeda's blueshift redshift z = -0.001, the Friedmann H(z) equals 69.969 km/s/Mpc = $2.269\times10^{?18}$ s$^{?1}$. Evaluated at the Hubble timescale t = t_H = 4.352×10^{17} s, the dimensionless Friedmann-UQFF Resonance Coefficient H_UQFF = H(z)×t_H ˜ **0.987** — a near-unity value representing gravitational doubling over cosmological time. This paper also introduces the M_visible/M_DM_mass cascade (explicit DM split tracking) and a minor ISM dust drag acceleration term. Together these constitute three additive UQFF 2.0 enhancements derived from the Andromeda module upgrade.

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1. Background

The UQFF framework computes total gravitational acceleration g_total(r,t) as a sum of Newtonian, Triadic (26-layer), cosmological, quantum, electromagnetic, fluid, resonant, and dark matter terms. PAPER_273–275 established the Andromeda-specific physics: blueshift approach amplifier ?, HI 21-cm resonance, and DM 80/20 NFW partition.

The original Andromeda implementation (pre-Session 76) lacked a time-dependent expansion coupling that accounts for the Friedmann-Lemaitre cosmological evolution of Hubble parameter H(z). This is significant because over a Hubble timescale, H(z)×t ˜ 1 — implying that expansion-mediated gravitational coupling is of the same order as the base Newtonian term.

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2. Friedmann H(z) Expansion Coupling

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The Hubble parameter as a function of redshift in the standard ?CDM cosmology:

$$H(z) = H_0\sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}$$

For Andromeda parameters: H0 = 70.0 km/s/Mpc, O_m = 0.3, O_? = 0.7, z = -0.001.

$$H(z = -0.001) = 70.0 \times \sqrt{0.3 \times (0.999)^3 + 0.7}$$

$$= 70.0 \times \sqrt{0.3 \times 0.997003 + 0.7} = 70.0 \times \sqrt{0.9991}$$

$$= 70.0 \times 0.99955 = 69.969 \text{ km/s/Mpc}$$

In SI units: H_SI = 69.969 × 10^3 / 3.086×10^{22} = **2.269×10?^{1}8 s?^{1}**

2.2 UQFF Expansion Term

The new UQFF gravity coupling term is:

$$g_{\text{expansion}}(r,t) = \frac{G \cdot M}{r^2} \cdot H(z) \cdot t$$

This represents the gravitational effect of cosmological expansion flow coupling into UQFF buoyancy at timescale t.

Numerical values for Andromeda: - g_base = G×M/r^2 = 6.674×10?^{11} × 1.989×104^2 / (1.04×10^{21})^2 = **1.227×10?^{1}° m/s^2** - At t = 1 Gyr (3.156×10^{1}6 s): g_expansion = 1.227×10?^{1}° × 2.269×10?^{1}8 × 3.156×10^{1}6 = **8.79×10?^{13} m/s^2** - At t = t_Hubble (4.352×10^{1}7 s): g_expansion = 1.227×10?^{1}° × 0.987 = **1.211×10?^{1}° m/s^2**

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3. H_UQFF Near-Unity Resonance Coefficient

3.1 Definition

$$H_{\text{UQFF}} = H(z) \times t_H$$

where t_H = 4.352×10^{1}7 s (Hubble time ˜ 13.8 Gyr).

For Andromeda (z = -0.001):

$$H_{\text{UQFF}} = 2.269 \times 10^{-18} \times 4.352 \times 10^{17} = \mathbf{0.987}$$

3.2 Near-Unity Discovery

H_UQFF ˜ 0.987 ˜ **1** is a remarkable result. It means:

> Over a Hubble timescale, the Friedmann expansion coupling adds an amount of gravitational acceleration equal to **98.7% of the base Newtonian term** — nearly doubling g_base.

This near-unity value is not coincidental. In a flat ?CDM universe (O_m + O_? = 1):

$$H_{\mathrm{UQFF}} = H_0 \times t_H \approx 1.0$$

because H0 × t_H is the dimensionless Hubble number ˜ 0.96–1.0 for observationally consistent H0.

New UQFF Constant: H_UQFF = H(z)×t_H — **Friedmann-UQFF Near-Unity Resonance Coefficient**

Parameter	Value
H0	70.0 km/s/Mpc
O_m	0.3
O_?	0.7
z (Andromeda)	-0.001
H(z)	2.269×10?¹8 s?¹
t_H	4.352×10¹7 s
H_UQFF	**0.987** (~1)
g_expansion(t_H)	1.211×10?¹° m/s²
g_base	1.227×10?¹° m/s²
Ratio g_exp/g_base	98.7%

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For blueshift z < 0: H(z) is slightly **lower** than H0 (matter term O_m×(1+z)³ is suppressed below O_m for z < 0), meaning:

$$H(z < 0) < H_0 \Rightarrow H_{\mathrm{UQFF}}(z < 0) < H_{\mathrm{UQFF}}(z = 0)$$

For Andromeda (z = -0.001): H_UQFF = 0.987 vs flat-universe H_UQFF0 = 0.9985. The blueshift suppresses H_UQFF by 0.15%.

This is the inverse of the redshift behaviour (z > 0 ? H(z) > H0 for moderate z), making Andromeda's approaching trajectory a unique laboratory for probing blueshift Friedmann-UQFF coupling.

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A second minor additive term captures ISM dust ram-pressure drag:

$$a_{\text{dust}} = \frac{\rho_{\text{dust}} \cdot v_{\text{orbit}}^2}{c^2 \cdot \rho_{\text{mean}}} \cdot g_{\text{base}}$$

where ?_mean = M/V_fluid = 1.989×104^2 / 106° = $1.989\times10?^{1}8$ kg/m^3.

Numerical value for Andromeda:

$$a_{\text{dust}} = \frac{10^{-20} \times (2.5 \times 10^5)^2}{(2.998 \times 10^8)^2 \times 1.989 \times 10^{-18}} \times 1.227 \times 10^{-10}$$

$$= \frac{6.25 \times 10^{-10}}{8.988 \times 10^{16} \times 1.989 \times 10^{-18}} \times 1.227 \times 10^{-10}$$

$$= \frac{6.25 \times 10^{-10}}{0.1788} \times 1.227 \times 10^{-10} = 4.29 \times 10^{-19} \text{ m/s}^2$$

At ~9 orders below g_base, the dust drag is physically negligible but dimensionally consistent and additive within the UQFF sum.

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When M is updated via `updateVariable("M", value)`, the derived quantities are now automatically recomputed:

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For Andromeda defaults: M = 1.989×104^2 kg, f_DM = 0.80: - M_visible = 0.20 × 1.989×104^2 = **3.978×104^1 kg** (20% visible baryons) - M_DM_mass = 0.80 × 1.989×104^2 = **1.591×104^2 kg** (80% dark matter)

These are tracked in `updateCache()` and exported via `exportState()` (params 26–32), providing explicit bookkeeping of the M31 baryonic/DM split consistent with PAPER_275.

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The full UQFF 2.0 equation for Andromeda with all PAPER_273–276 terms:

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Term magnitudes at t=0 (current epoch):

Term	Value (m/s^2)	Paper
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Ug_sum (26 layers)	6.380×10??	26-layer Triadic
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F_res(?_HI) at t=0	1×10?^{12}	PAPER_274
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Note: g_expansion grows linearly with t, dominating over g_grav at t = t_H.

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Seven new parameters added to `exportState()`: 1. `H0` = 70.0 km/s/Mpc 2. `Omega_m` = 0.3 3. `Omega_Lam` = 0.7 4. `Mpc_to_m` = 3.086×10^{22} 5. `rho_dust` = 1×10?$^{2\circ}$ kg/m^3 6. `M_visible` = (1–f_DM)×M 7. `M_DM_mass` = f_DM×M

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Three new UQFF enhancements for Andromeda (PAPER_276):

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$$= \frac{6.25 \times 10^{-10}}{0.1788} \times 1.227 \times 10^{-10} = 4.29 \times 10^{-19} \text{ m/s}^2$$

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Term magnitudes at t=0 (current epoch):

Term	Value (m/s^2)	Paper
g_grav = G·M/r^2	1.227×10?$^{1°}$	baseline
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?-term	2.998×10?^{3}6	cosmological
g_quantum	~2×10?^{2}6	HUP
g_Lorentz	~2×10?^{3}6	IGM EM
g_fluid	~8×10?^{23}	IGM buoyancy
F_res(?_HI) at t=0	1×10?^{12}	PAPER_274
g_DM	1.356×10?$^{1°}$	PAPER_275
g_expansion at t=0	0	PAPER_276
a_dust	4.29×10?^{1}?	PAPER_276
?_approach	1.001001	PAPER_273

Note: g_expansion grows linearly with t, dominating over g_grav at t = t_H.

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7. exportState Update (25 ? 32 parameters)

Seven new parameters added to `exportState()`: 1. `H0` = 70.0 km/s/Mpc 2. `Omega_m` = 0.3 3. `Omega_Lam` = 0.7 4. `Mpc_to_m` = 3.086×10^{22} 5. `rho_dust` = 1×10?$^{2°}$ kg/m^3 6. `M_visible` = (1–f_DM)×M 7. `M_DM_mass` = f_DM×M

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8. Conclusions

Three new UQFF enhancements for Andromeda (PAPER_276):

1. **Friedmann-UQFF Expansion Coupling**: g_expansion = g_base × H(z) × t. At t = t_H, adds 98.7% of g_base — near-gravitational doubling over cosmological time.
2. **H_UQFF = 0.987**: New UQFF constant (Friedmann-UQFF Near-Unity Resonance Coefficient). Near-unity reflects the fundamental relationship H0 × t_H ˜ 1 in flat ?CDM. Andromeda's blueshift (z = -0.001) makes H_UQFF 0.15% below flat-universe value — a blueshift Friedmann suppression.
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Andromeda Friedmann-UQFF Gravity Coupling: H(z)×t Expansion Term and H_UQFF Near-Unity Resonance

Author: Daniel T. Murphy
Session: 76 (March 2026)
Module: ANDROMEDA_UQFF_MODULE.cpp (M31 Master, UQFF 2.0)
WOLFRAM_TERM: `AndromedaUQFF:g_exp=G*M/r^2*H(z)*t; H(z)=H0*Sqrt[Om_m*(1+z)^3+Om_L]/M`
`H_UQFF=H(z)*t_H~0.987 [PAPER_276]`

Abstract

This paper introduces a new UQFF gravity term for the Andromeda Galaxy (M31): the **Friedmann-UQFF Expansion Coupling**, defined as g_expansion = ($G \cdot M/r^2$) × H(z) × t. For Andromeda's blueshift redshift z = -0.001, the Friedmann H(z) equals 69.969 km/s/Mpc = 2.269×10?^{1}8 s?^{1}. Evaluated at the Hubble timescale t = t_H = 4.352×10^{1}7 s, the dimensionless Friedmann-UQFF Resonance Coefficient H_UQFF = H(z)×t_H ˜ **0.987** — a near-unity value representing gravitational doubling over cosmological time. This paper also introduces the M_visible/M_DM_mass cascade (explicit DM split tracking) and a minor ISM dust drag acceleration term. Together these constitute three additive UQFF 2.0 enhancements derived from the Andromeda module upgrade.

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1. Background

The UQFF framework computes total gravitational acceleration g_total(r,t) as a sum of Newtonian, Triadic (26-layer), cosmological, quantum, electromagnetic, fluid, resonant, and dark matter terms. PAPER_273–275 established the Andromeda-specific physics: blueshift approach amplifier ?, HI 21-cm resonance, and DM 80/20 NFW partition.

The original Andromeda implementation (pre-Session 76) lacked a time-dependent expansion coupling that accounts for the Friedmann-Lemaitre cosmological evolution of Hubble parameter H(z). This is significant because over a Hubble timescale, H(z)×t ˜ 1 — implying that expansion-mediated gravitational coupling is of the same order as the base Newtonian term.

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2. Friedmann H(z) Expansion Coupling

2.1 Friedmann Equation

The Hubble parameter as a function of redshift in the standard ?CDM cosmology:

$$H(z) = H_0\sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}$$

For Andromeda parameters: H0 = 70.0 km/s/Mpc, O_m = 0.3, O_? = 0.7, z = -0.001.

$$H(z=-0.001) = 70.0 \times \sqrt{0.3 \times (0.999)^3 + 0.7}$$

$$= 70.0 \times \sqrt{0.3 \times 0.997003 + 0.7} = 70.0 \times \sqrt{0.9991}$$

$$= 70.0 \times 0.99955 = 69.969 \text{ km/s/Mpc}$$

In SI units: H_SI = 69.969 × 10^3 / 3.086×10^{22} = **2.269×10?^{1}8 s?^{1}**

2.2 UQFF Expansion Term

The new UQFF gravity coupling term is:

$$g_{\text{expansion}}(r,t) = \frac{G \cdot M}{r^2} \cdot H(z) \cdot t$$

This represents the gravitational effect of cosmological expansion flow coupling into UQFF buoyancy at timescale t.

Numerical values for Andromeda: - g_base = G×M/r^2 = 6.674×10?^{11} × 1.989×104^2 / (1.04×10^{21})^2 = **1.227×10?^{1}° m/s^2** - At t = 1 Gyr (3.156×10^{1}6 s): g_expansion = 1.227×10?^{1}° × 2.269×10?^{1}8 × 3.156×10^{1}6 = **8.79×10?^{13} m/s^2** - At t = t_Hubble (4.352×10^{1}7 s): g_expansion = 1.227×10?^{1}° × 0.987 = **1.211×10?^{1}° m/s^2**

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3. H_UQFF Near-Unity Resonance Coefficient

3.1 Definition

$$H_{\text{UQFF}} = H(z) \times t_H$$

where t_H = 4.352×10^{1}7 s (Hubble time ˜ 13.8 Gyr).

For Andromeda (z = -0.001):

$$H_{\text{UQFF}} = 2.269 \times 10^{-18} \times 4.352 \times 10^{17} = \mathbf{0.987}$$

3.2 Near-Unity Discovery

H_UQFF ˜ 0.987 ˜ **1** is a remarkable result. It means:

> Over a Hubble timescale, the Friedmann expansion coupling adds an amount of gravitational acceleration equal to **98.7% of the base Newtonian term** — nearly doubling g_base.

This near-unity value is not coincidental. In a flat ?CDM universe (O_m + O_? = 1):

$$H_{\mathrm{UQFF}} = H_0 \times t_H \approx 1.0$$

because H0 × t_H is the dimensionless Hubble number ˜ 0.96–1.0 for observationally consistent H0.

New UQFF Constant: H_UQFF = H(z)×t_H — **Friedmann-UQFF Near-Unity Resonance Coefficient**

Parameter	Value
H0	70.0 km/s/Mpc
O_m	0.3
O_?	0.7
z (Andromeda)	-0.001
H(z)	2.269×10?¹8 s?¹
t_H	4.352×10¹7 s
H_UQFF	**0.987** (~1)
g_expansion(t_H)	1.211×10?¹° m/s²
g_base	1.227×10?¹° m/s²
Ratio g_exp/g_base	98.7%

3.3 Blueshift Sensitivity

For blueshift z < 0: H(z) is slightly **lower** than H0 (matter term O_m×(1+z)³ is suppressed below O_m for z < 0), meaning:

$$H(z < 0) < H_0 \Rightarrow H_{\mathrm{UQFF}}(z < 0) < H_{\mathrm{UQFF}}(z = 0)$$

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UQFF computed: MUGE buoyancy ratio U_bi/F_U = [SSq]×?×r^2/GM = 5.7e-1×5.0e-4 = 2.85e-4; compressed MUGE baseline g = 5.4e-7 m/s^2 at r_ISCO.